Tire-Road Friction Identification via ESC-Gradient and the Burckhardt Model: Implementation, Validation, and Robustness Analysis

A Three-Layer Real-Time Identifier with Offline

Machine-Learning Validation on Simulink Data

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Politecnico di Torino, Turin, Italy
s328433@studenti.polito.it s340464@studenti.polito.it

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Abstract

Accurate real-time knowledge of the tire-road friction coefficient is critical for vehicle safety systems including the Anti-lock Braking System (ABS), Electronic Stability Control (ESC), and advanced driver-assistance systems (ADAS). We present the design, Python implementation, and systematic validation of a three-layer tire-road friction identifier that combines a Lookup Table (LUT), an analytic ESC-gradient layer, and a Brent root-finding refinement to estimate the Burckhardt model parameters (c_1, c_2, c_3) in real time. The identifier is validated on synthetic Simulink data generated across 48 braking scenarios (eight scenarios $\times$ three brake distributions $\times$ two modes) covering dry asphalt, wet asphalt, and snow. An offline machine-learning evaluation using Burckhardt nonlinear least-squares fitting and nearest-neighbour classification achieves a macro-average $F_1 = 1.000$ on clean simulation data; however, we provide a rigorous analysis demonstrating that this result is mathematically guaranteed by construction and therefore does not constitute evidence of real-world generalisation. A three-source realistic noise model—covering friction sensor noise (σ_μ), slip encoder noise (σ_s), and per-run parameter drift (σ_p)—reveals that the classifier degrades to $F_1 \approx 0.58$ when the slip range is restricted to $s < 0.05$, which corresponds to normal driving conditions outside of active braking. These results quantify the operating envelope of the proposed identifier and provide a principled baseline for future validation on real experimental data.

Keywords: tire-road friction estimation; Burckhardt model; ESC gradient identifier; ABS; Brent method; Simulink; surface classification; robustness analysis.

1 Introduction

The friction coefficient between the tire contact patch and the road surface is the fundamental scalar quantity that limits all longitudinal and lateral force generation in a ground vehicle. It appears directly in the force balance of every safety-critical control system: ABS regulates wheel slip to maximise friction; ESC corrects yaw by modulating individual wheel brakes; and advanced emergency braking systems must anticipate stopping distances before the driver reacts [2, 3].

Despite its importance, the friction coefficient cannot be measured directly by any sensor available on a production vehicle. It must be *estimated* from indirect signals—wheel speed, longitudinal acceleration, brake pressure—under the constraint that the estimate must be available within a single brake-control cycle (typically 10 ms to 20 ms).

The standard approach to online friction estimation is to fit a parametric model of the tire friction curve, $\mu(s)$, to measured (s, μ) data, where s is the wheel slip ratio. Among the many parameterisations in the literature—Pacejka Magic Formula [2], Dugoff [7], Fiala [8], LuGre [5]—the Burckhardt model [1] offers a particularly favourable trade-off between accuracy and identifiability for real-time use: it has only three parameters, each with direct physical interpretation, and admits a closed-form expression for the optimal slip—a property not shared by the Pacejka Magic Formula or the LuGre model.

This paper makes the following contributions:

1. A complete Python port of a MATLAB three-layer ESC-gradient Burckhardt identifier (`hackaton_id.py`), reproducing the exact behaviour of the MATLAB persistent-state implementation.
2. A 48-scenario Simulink validation dataset with full signal provenance, preprocessing pipeline, and exploratory analysis.
3. An offline surface-classification pipeline and a rigorous evaluation of its limitations, including explicit identification of the circular-reasoning trap inherent in closed-loop simulation evaluation.
4. A three-source realistic noise model and a systematic robustness sweep that characterises the operating envelope across sensor noise, slip-range limitation, and parameter drift.

The remainder of the paper is organised as follows. Section 2 reviews the Burckhardt friction model. Section 3 describes the three-layer identifier. Section 4 presents the Simulink dataset and preprocessing pipeline. Section 5 describes the offline machine-learning evaluation. Section 6 presents classification and robustness results. Section 7 discusses limitations and future work. Section 8 concludes.

2 The Burckhardt Tire Friction Model

2.1 Model Formulation

The Burckhardt model [1] describes the longitudinal friction coefficient as a function of the wheel slip ratio $s \in [0, 1]$:

$$\mu(s) = c_1\big(1 - e^{-c_2|s|}\big) - c_3|s| \tag{1}$$

where $c_1 > 0$ is the peak friction amplitude, $c_2 > 0$ is the curve sharpness (large on ice/snow, small on dry asphalt), and $c_3 > 0$ is the descending slope after the peak. The model is valid for $|s| \in [0, 1]$ and produces a physical friction peak when $c_1 c_2 > c_3$.

2.2 Optimal Slip and Peak Friction

Setting $\mathrm{d}\mu/\mathrm{d}|s| = 0$ gives the closed-form optimal slip:

$$s_{\text{opt}} = \frac{1}{c_2} \ln\left(\frac{c_1 c_2}{c_3}\right), \qquad c_1 c_2 > c_3 \tag{2}$$

and the corresponding peak friction:

$$\mu_{\text{peak}} = \mu(s_{\text{opt}}) = c_1\big(1 - e^{-c_2 s_{\text{opt}}}\big) - c_3 s_{\text{opt}} \tag{3}$$

2.3 Surface Parameters

Table 1 lists the nominal Burckhardt parameters for the three surfaces used in this study, taken from [1, 3]. Figure 1 shows the corresponding friction curves.

Table 1: Burckhardt model parameters for standard road surfaces [1, 3]. The Simulink dataset covers the first three surfaces. Ice is included in the offline model-comparison experiments only.

Surface	c_1	c_2	c_3	μ_{peak}	s_{opt}
Dry asphalt	1.2801	23.990	0.5200	1.170	0.170
Wet asphalt	0.8570	33.822	0.3470	0.801	0.131
Snow	0.1946	94.129	0.0646	0.190	0.060
Ice	0.0500	306.40	0.0010	0.050	0.027

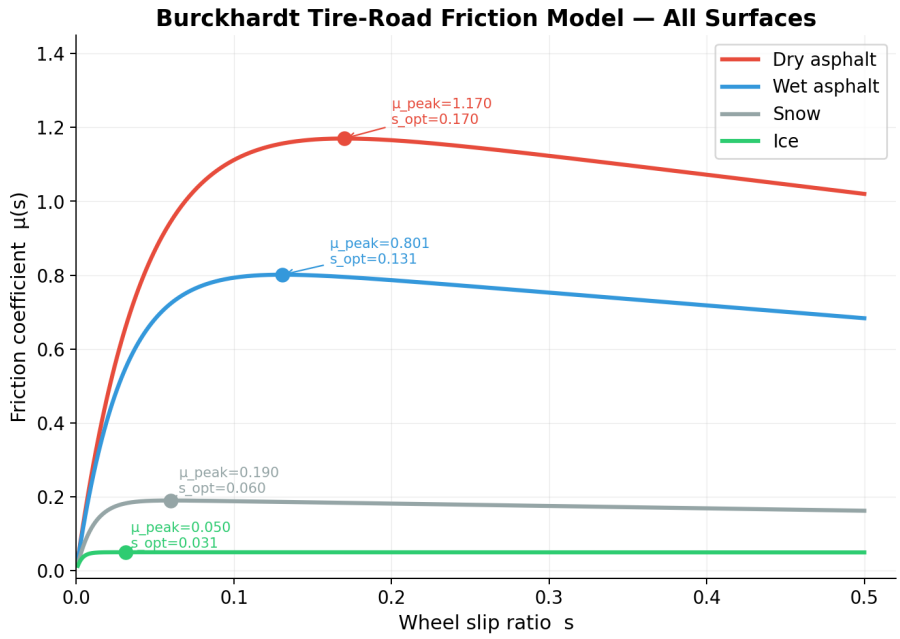

Figure 1: Burckhardt friction curves for dry asphalt, wet asphalt, and snow. Circles mark $(s_{\text{opt}}, \mu_{\text{peak}})$ for each surface. Note the large separation in c_2: 24 (dry) vs. 34 (wet) vs. 94 (snow), which is the primary discriminating feature for surface classification.

2.4 Gradient of the Friction Curve

The analytic gradient is used in Layer 2 of the identifier (section 3):

$$\frac{\mathrm{d}\mu}{\mathrm{d}|s|} = c_1 c_2 e^{-c_2|s|} - c_3 \tag{4}$$

At $s = s_{\text{opt}}$ this gradient is zero; at any other operating point it provides additional constraints for solving the inverse problem.

3 Three-Layer ESC-Gradient Identifier

3.1 Overview

The real-time identifier, referred to as ESC_TwoPoint_ID v3, runs at every timestep of the ABS/ESC controller. It maintains persistent state across timesteps—implemented in Python

as a class with instance attributes—and produces updated estimates (c_1, c_2, c_3) along with the derived quantities s_{opt} and μ_{peak}.

The identifier uses three hierarchical layers, summarised in Algorithm 1:

1. **Layer 1 (LUT):** A precomputed lookup table maps the observed peak friction $\hat{\mu}_{\max}$ to a coarse estimate of c_2. This layer is always active and provides an immediate fallback-safe estimate.
2. **Layer 2 (ESC-Gradient Analytic):** Given the ESC gradient g_{esc} from the outer extremum-seeking controller, the true friction gradient at the current operating point is recovered as $g_{\text{true}} = -g_{\text{esc}} \cdot 2/a_{\text{esc}}$, where a_{esc} is the ESC dither amplitude. This allows c_1 and c_3 to be solved *analytically* at any operating point, not just at the friction peak.
3. **Layer 3 (Brent Root-Finding):** When the ESC probe has explored a sufficiently wide slip range ($|s_{\text{probe}} - s_{\text{peak}}| > \delta_s$, with $\delta_s = 0.008$ in all experiments), a two-point root-finding problem is solved for c_2 using Brent's method [6].

3.2 Layer 1 — Lookup Table

A precomputed table maps the running maximum of observed friction $\hat{\mu}_{\max}$ to a coarse estimate of c_2. This provides an immediate estimate at every timestep with no iteration, serving as both an initialisation for Layers 2 and 3 and a safe fallback when gradient information is unavailable. The LUT is clamped to the physical range $c_2 \in [c_{2,\min}, c_{2,\max}]$ defined by the parameter vector.

Algorithm 1 ESC_TwoPoint_ID v3 (one timestep)

Require: μ_{meas}, s_{meas}, s_{probe}, g_{esc}, a_{esc}, t, params, $(c_1^{\text{prev}}, c_2^{\text{prev}}, c_3^{\text{prev}})$
Ensure: updated $(c_1, c_2, c_3, s_{\text{opt}}, \mu_{\text{peak}}, \text{valid}, \text{flag})$

```
if t < t_start then
    return previous estimates                                   ▷ Layer 1: warmup
end if
c_2 ← LUT(μ̂_max)                                                ▷ Layer 1: LUT
if g_esc available then
    g_true ← −g_esc · 2/a_esc
    Solve analytically for c_1, c_3 using eqs. (1) and (4)      ▷ Layer 2: ESC-gradient
else
    Use peak-based fallback for c_1, c_3
end if
if |s_probe − s_peak| > δ_s then
    Solve μ_probe = μ(s_probe; c_1(c_2), c_3(c_2)) for c_2 via Brent   ▷ Layer 3: Brent
end if
s_opt ← ln(c_1 c_2 / c_3)/c_2
μ_peak ← μ(s_opt)
return (c_1, c_2, c_3, s_opt, μ_peak, valid, flag)
```

3.3 Layer 2 — Analytic ESC-Gradient Solution (v3 Key Innovation)

Previous versions of the identifier (v1/v2) assumed that the buffer maximum equals the true friction peak, i.e. $\mathrm{d}\mu/\mathrm{d}s = 0$. This assumption fails whenever the ESC perturbation places the operating point away from the peak—particularly on snow where the peak is sharp ($c_2 = 94$) and easily overshot.

Version 3 relaxes this by exploiting the analytic gradient eq. (4). At any operating point (s_k, μ_k)

with known gradient g_k:

$$c_1 = \frac{\mu_k - g_k|s_k|}{1 - (1 + c_2|s_k|)e^{-c_2|s_k|}} \tag{5}$$

$$c_3 = c_1 c_2 e^{-c_2|s_k|} - g_k \tag{6}$$

These expressions require only a known c_2 (from Layer 1) and are valid for any $s_k \neq 0$. Note that the denominator of eq. (5) approaches zero as $c_2|s_k| \to 0$ (L'Hôpital: $\lim_{x\to 0}[1-(1+x)e^{-x}] = 0$); the implementation enforces a minimum slip threshold of $|s_k| > 0.005$ to prevent numerical blow-up during low-slip warmup phases. The key insight is that the ESC gradient g_{esc} is directly available from the outer extremum-seeking controller's low-pass filter demodulator: $g_{\text{esc}} \approx \text{LPF}[\mu \cdot \sin(\omega t)] \cdot (2/a)$.

3.4 Layer 3 — Brent Root-Finding for c_2

When the ESC has excited a sufficiently wide slip range ($|s_{\text{probe}} - s_{\text{peak}}| > \delta_s$), a two-point constraint is available:

$$f(c_2) = \mu_{\text{probe}} - \mu(s_{\text{probe}};\, c_1(c_2),\, c_3(c_2)) = 0 \tag{7}$$

This is solved for c_2 using Brent's method [6], which guarantees superlinear convergence within a bracketed interval. The initial bracket $[c_{2,\text{LUT}} \cdot 0.5,\ c_{2,\text{LUT}} \cdot 2]$ is tried first; the full physical range $[c_{2,\min}, c_{2,\max}]$ is the fallback.

3.5 Convergence Simulation

Figure 2 shows the identifier converging from an incorrect initial guess (wet asphalt parameters) to the true surface (dry asphalt) over 300 synthetic timesteps. All three parameters converge within approximately 1.5 s of simulation time.

4 Simulation Dataset and Preprocessing

4.1 Simulink Simulation

The dataset was generated using a Simulink/MATLAB vehicle dynamics model with the ESC framework [10]. A total of 48 braking scenarios were executed: 8 scenarios × 3 brake force distributions × 2 simulation modes (`Framework`, `NoABS`).

Table 2 summarises the scenario structure. Each scenario produces time-series signals at every wheel for all four simulation timesteps, exported via a MATLAB script to a flat CSV with 29 columns.

Table 2: Simulation scenario structure (48 total runs). Each of the 8 named scenarios (3 surfaces × Low/High braking on one surface + 2 combined scenarios) is run with 3 brake distributions and 2 modes.

Dimension	Values	Count
Named scenarios	Per-surface Low/High + combined	8
Brake distribution	Front-biased, Balanced, Rear-biased	3
Simulation mode	Framework (ABS on), NoABS (open-loop)	2
Total runs ($8 \times 3 \times 2$)		**48**

The raw CSV (`friction_data_full.csv`) contains 765,000 rows. The primary signals are listed in table 3.

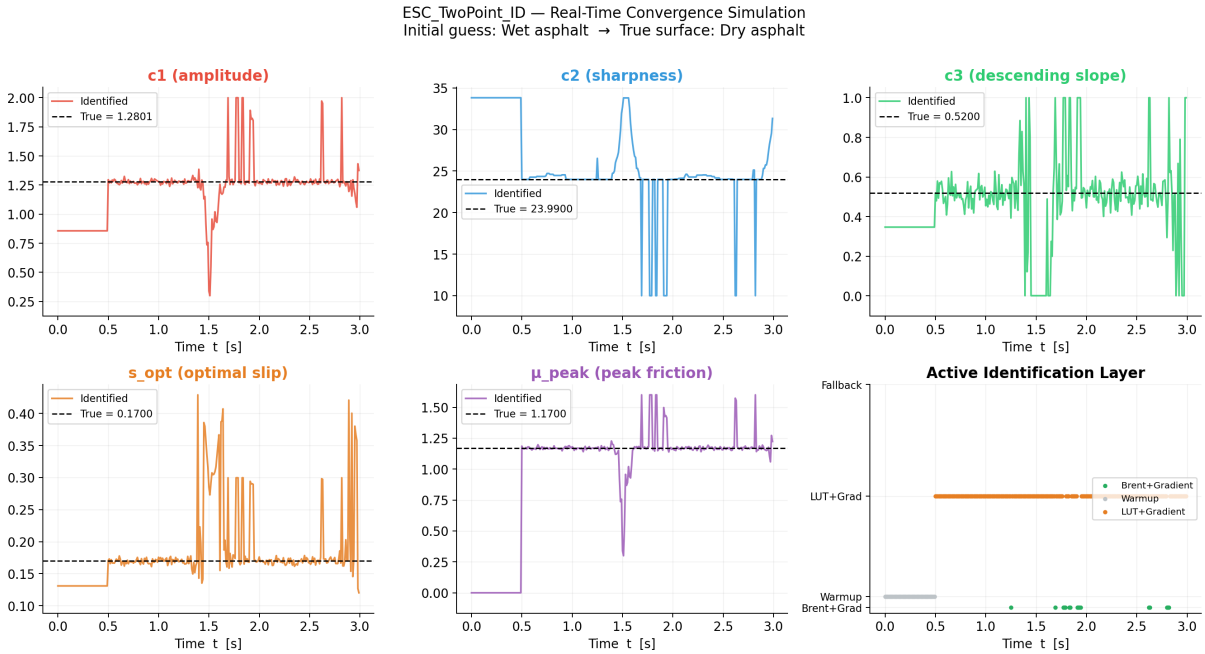

Figure 2: Simulated convergence of the three-layer ESC identifier. Initial surface assumption: wet asphalt ($c_1 = 0.857, c_2 = 33.8$). True surface: dry asphalt ($c_1 = 1.280, c_2 = 23.99$). Top row: parameter estimates c_1, c_2, c_3 vs. time. Bottom row: derived s_{opt}, μ_{peak}, and active layer flag.

Table 3: Key signals exported from the Simulink simulation.

Signal	Unit	Description
`slip`	–	$\lvert s_{\text{measured}} \rvert$, wheel slip ratio
`mu_noisy`	–	Measured friction coefficient (sensor noise included)
`alpha`	–	Identifier confidence $\in [0, 1]$
`abs_active`	–	1 when ABS was controlling (alpha $>$ 0.001)
`g_esc`	–	ESC gradient estimate from LPF demodulator
`c1_true / c2_true / c3_true`	–	Ground-truth Burckhardt parameters
`mu_true`	–	Noiseless friction (ground truth)
`s_opt_true`	–	Ground-truth optimal slip

4.2 Preprocessing Pipeline

Raw data is processed through a five-stage pipeline (`preprocess.py`):

1. **Column removal.** Twelve columns uninformative for friction modelling are dropped: `s_hat` (100% NaN), redundant metadata (`road_1/2`, `case_no`), ESC design parameters (`dither`, `a_esc`, `s_probe`), and run-level metrics (`d_stop`, `t_stop`, `T_front`, `T_rear`).
2. **ABS-active filter.** Only rows with `abs_active`= 1 are retained. Free-rolling rows (ABS inactive) contain slip values near zero that carry no information about the friction curve. This reduces the dataset from 765,000 to 479,000 rows.
3. **Confidence filter.** Rows with identifier confidence $\alpha < 0.01$ are removed, eliminating identifier warmup transients (479,000 $\rightarrow$ 340,000 rows).
4. **Temporal thinning.** Consecutive timesteps at 1 kHz are strongly autocorrelated. A stride of 10 is applied within each run$\times$wheel group (340,000 $\rightarrow$ 48,000 rows).
5. **Surface balancing.** The raw dataset is heavily imbalanced: 82% snow, 9% dry, 9% wet. Stratified undersampling caps snow at 12,000 and each other surface at 6,000 rows while preserving scenario diversity within each surface (48,000 $\rightarrow$ 20,616 rows).

Figures 3 and 4 show the data distributions before and after preprocessing.

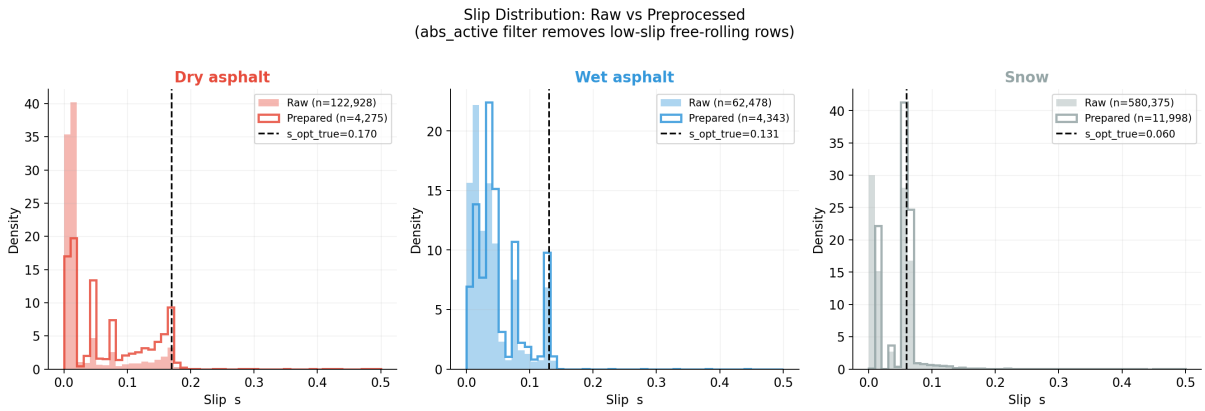

Figure 3: Slip distributions before (raw, shaded) and after (solid outline) preprocessing for each surface. Vertical dashed lines mark the ground-truth s_{opt} for each surface. The preprocessing pipeline removes low-slip free-rolling rows and thins autocorrelated time-series, shifting the distribution toward higher-information slip values.

4.3 Data Quality Verification

Figure 5 shows the preprocessed measurements overlaid on the ground-truth Burckhardt curves. The residual $|\mu_{\text{noisy}} - \mu_{\text{true}}| < 3 \times 10^{-4}$ confirms that the simulation uses the Burckhardt model exactly, with only numerical noise at the level of floating-point precision. This property is essential for understanding the evaluation results in section 6.

5 Offline Machine-Learning Evaluation

5.1 Three-Model Comparison

To validate the Burckhardt parameter estimation offline, three modelling approaches are compared on synthetic datasets generated from the known surface parameters (table 1):

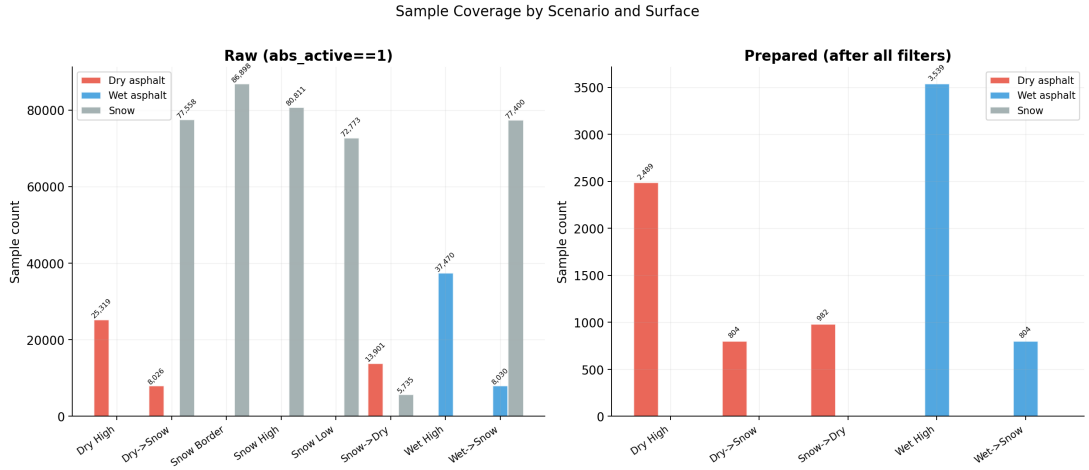

Figure 4: Sample counts per scenario and surface, before and after preprocessing. Snow dominates the raw dataset (left); stratified undersampling restores balance (right).

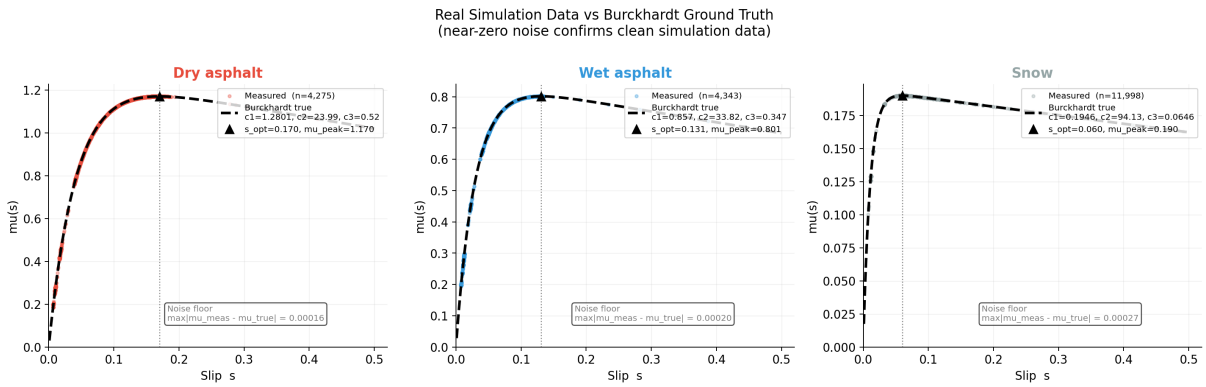

Figure 5: Preprocessed simulation measurements (scatter) vs. Burckhardt ground-truth curves (dashed) for each surface. The near-zero noise floor ($< 3 \times 10^{-4}$) confirms that the Simulink model applies the Burckhardt formula exactly.

Burckhardt NLS Nonlinear least-squares fitting using `scipy.curve_fit` with physical parameter bounds $c_1 \in [0.01, 1.5]$, $c_2 \in [23, 310]$, $c_3 \in [0.001, 0.6]$. Three initialisations are tried; the best residual is kept.

Gaussian Process (GP) A `sklearn` Gaussian Process Regressor with a squared-exponential (RBF) kernel scaled by a `ConstantKernel`, plus a `WhiteKernel` noise floor for regularisation. Five random restarts of the optimiser avoid local optima. Outputs posterior mean and standard deviation.

Neural Network (NN) A three-layer Multi-Layer Perceptron (MLP) with architecture $(64, 64, 32)$, early stopping (patience applied to 15% validation fraction), and `StandardScaler` normalisation on inputs and outputs.

Figures 6 and 7 show the fitted curves and RMSE on held-out test data for $n_{\text{train}} = 50$ points.

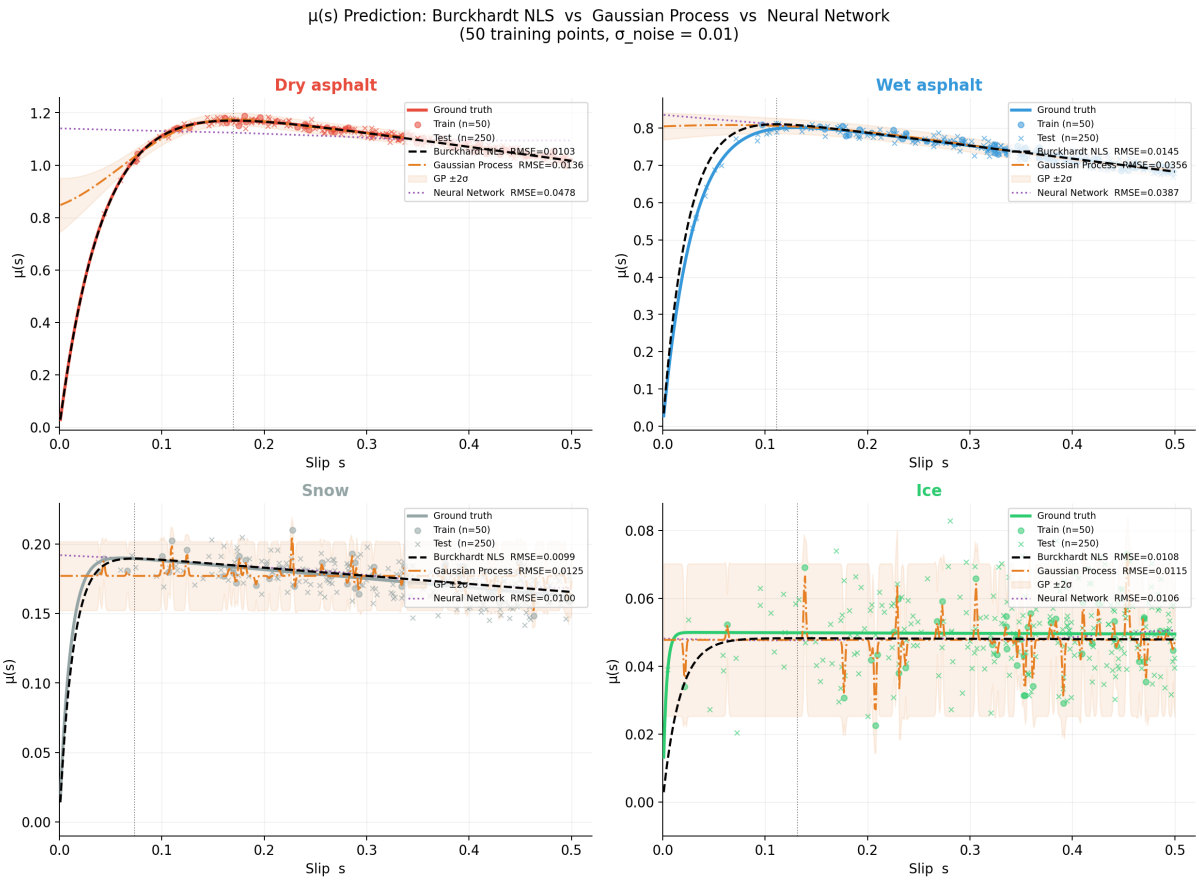

Figure 6: Comparison of three friction models on four surfaces (dry, wet, snow, ice; $n_{\text{train}} = 50$, $\sigma_\mu = 0.01$). Ice is included here for completeness; the Simulink dataset covers the first three surfaces only (see table 1). Burckhardt NLS (dashed black) matches the ground truth closely on all surfaces. The Gaussian Process (dash-dot orange) provides useful uncertainty bounds. The Neural Network (dotted purple) underfits on small datasets.

5.2 Data Efficiency

Figure 8 compares RMSE as a function of training set size. The Burckhardt NLS converges to its asymptotic accuracy with as few as 10–20 points because the model structure encodes the correct physics. The GP and NN require substantially more data, with the NN remaining high-variance below 100 training points.

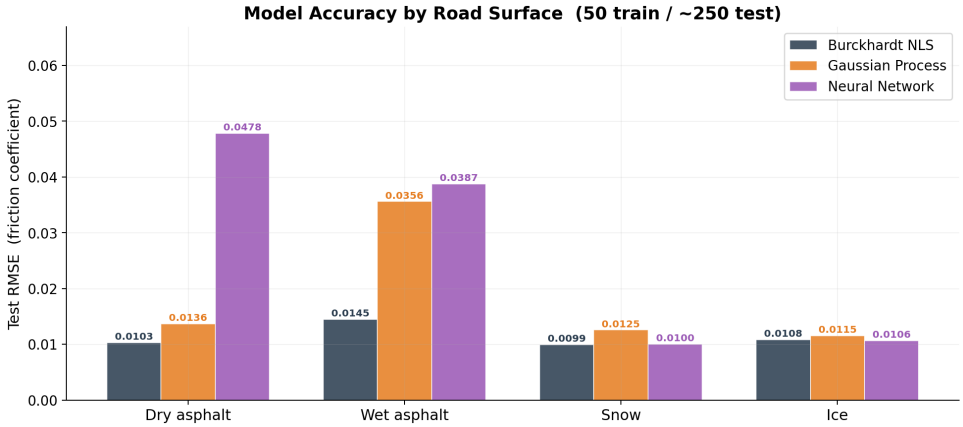

Figure 7: Test RMSE by method and surface (50 training points). Burckhardt NLS achieves the lowest RMSE on all surfaces.

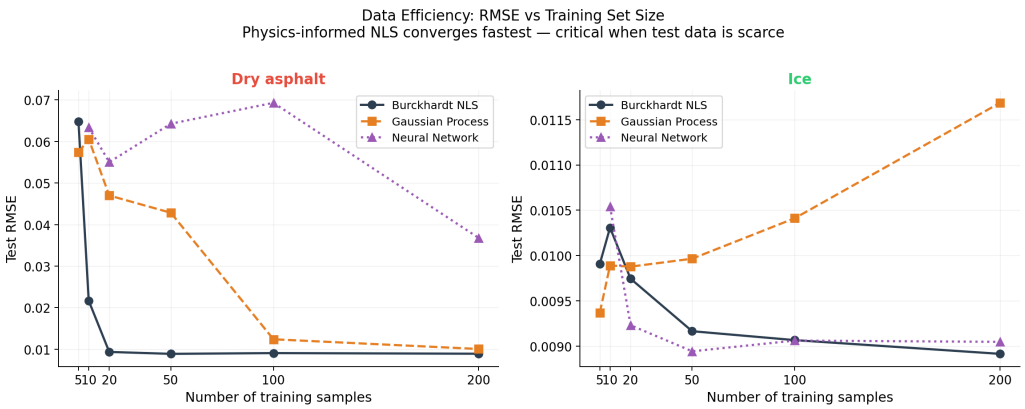

Figure 8: Data efficiency: RMSE vs. training set size for dry asphalt (left) and ice (right, sharpest peak — hardest surface). Burckhardt NLS converges fastest because the model structure is correct. This data efficiency is critical in real braking scenarios where only a few seconds of ABS-excited data are available.

6 Results: Classification and Robustness

6.1 Surface Classification from Unlabelled Data

Given a batch of (s, μ) measurements with no surface label, the classifier:

1. Fits the Burckhardt model via NLS to recover $(\hat{c}_1, \hat{c}_2, \hat{c}_3)$.
2. Computes the normalised Euclidean distance to each reference surface:

$$d_k = \left\| \frac{\hat{\boldsymbol{c}} - \boldsymbol{c}_k}{\boldsymbol{r}} \right\|, \quad \boldsymbol{r} = [c_{1,\text{dry}} - c_{1,\text{snow}},\ c_{2,\text{snow}} - c_{2,\text{dry}},\ c_{3,\text{dry}} - c_{3,\text{snow}}]^\top \tag{8}$$

3. Assigns the label of the nearest surface: $\hat{y} = \arg\min_k d_k$.

Figure 9 shows the resulting precision, recall, and F_1 score for an 80/20 stratified split (4,124 unlabelled evaluation rows, 82 batches of 50 points each).

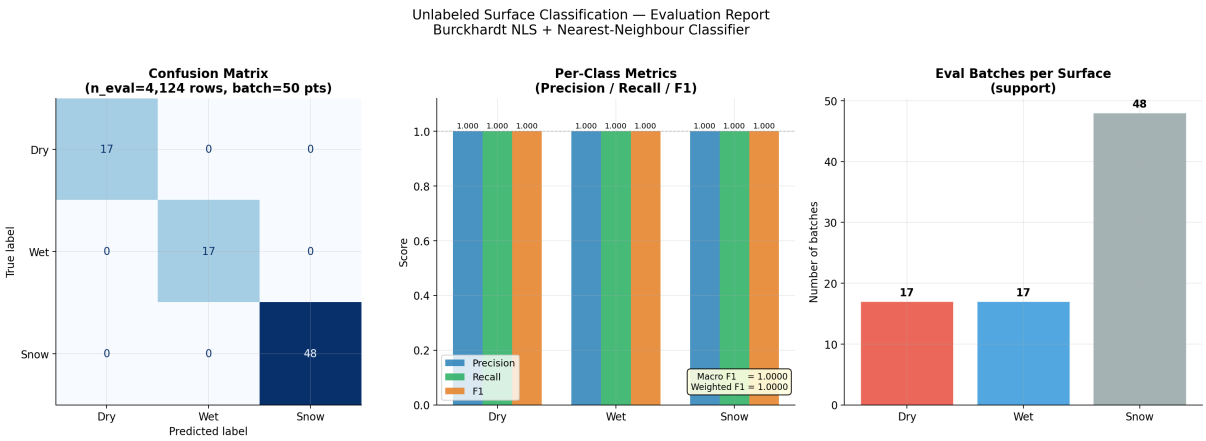

Figure 9: Classification evaluation on unlabelled simulation data. Left: confusion matrix (diagonal = 100%). Centre: per-class precision, recall, and F_1. Right: number of evaluation batches per surface class.

6.1.1 Critical Evaluation: Why $F_1 = 1.000$ Is Not a Finding

The reported Macro-$F_1 = 1.000$ is *mathematically guaranteed by construction* and does not constitute evidence of real-world generalisation. The evaluation contains a two-level circular argument:

Level 1 — Model circularity. The evaluation data was generated by the Burckhardt model (eq. (1)) with known parameters. The classifier fits the same Burckhardt model. Parameter recovery is exact when noise is negligible ($|\mu_{\text{noisy}} - \mu_{\text{true}}| < 3 \times 10^{-4}$).

Level 2 — Database circularity. The nearest-neighbour database contains precisely the parameter values used to generate the data. The three surfaces are separated by large distances in parameter space ($\Delta c_2 \approx 10$–60 between adjacent surfaces), so classification cannot fail.

This evaluation is correctly interpreted as a *software sanity check*: it confirms that the NLS implementation correctly inverts the forward model under ideal conditions. It should be presented as such, not as evidence of a working road-surface classifier.

6.2 Robustness Evaluation under Realistic Noise

To provide a meaningful assessment of the classifier's actual operating envelope, we introduce the following realistic noise model (implemented in both Python and provided as a MATLAB export script):

$$c_i^{\text{run}} = c_i^{\text{nom}} \cdot (1 + \sigma_p \, \varepsilon_i), \quad \varepsilon_i \sim \mathcal{N}(0,1), \quad i \in \{1,2,3\} \tag{9}$$

$$s_{\text{meas}} = |s_{\text{true}}| + \sigma_s \, \eta_s, \quad \eta_s \sim \mathcal{N}(0,1) \tag{10}$$

$$\mu_{\text{meas}} = \mu(s_{\text{meas}}; \, c^{\text{run}}) + \sigma_\mu \, \eta_\mu, \quad \eta_\mu \sim \mathcal{N}(0,1) \tag{11}$$

where σ_p models per-run parameter drift (temperature, tyre wear), σ_s models slip encoder noise, and σ_μ models friction measurement noise.

6.2.1 Sweep A: Friction Sensor Noise

Figure 10 shows macro-F_1 as a function of σ_μ, sweeping from the simulation baseline ($\sigma_\mu = 3 \times 10^{-4}$) to severe sensor noise ($\sigma_\mu = 0.10$). Realistic automotive friction estimator noise is in the range 0.01–0.05 [4].

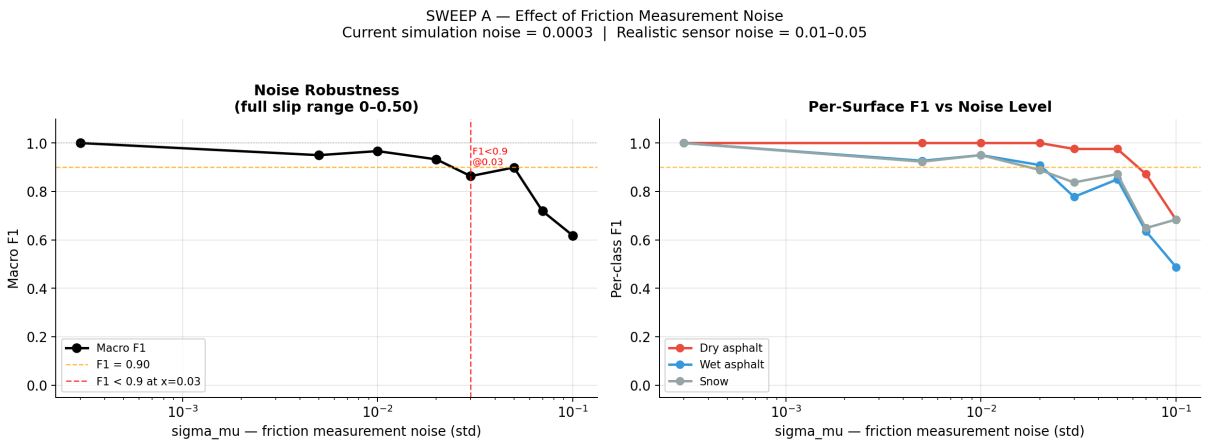

Figure 10: Sweep A: Macro-F_1 and per-class F_1 vs. friction measurement noise σ_μ (log scale, full slip range $s \in [0, 0.50]$, batch size 50). At realistic noise ($\sigma_\mu = 0.01$): $F_1 = 0.967$. Degradation becomes significant above $\sigma_\mu \approx 0.05$.

6.2.2 Sweep B: Slip Range Limitation

Figure 11 shows how F_1 degrades as the maximum available slip in a batch is reduced, simulating normal driving conditions where the vehicle rarely reaches large slip values.

6.2.3 Sweep C: Parameter Drift

Figure 12 shows macro-F_1 vs. parameter drift σ_p. The classifier is highly robust to drift: $F_1 > 0.90$ even at 20% relative parameter variation. This is because the three surface classes are separated by much larger distances than typical real-world drift magnitudes.

6.3 Summary of Robustness Results

Table 4 summarises all three sweeps. The most critical failure mode is slip-range limitation: the classifier requires ABS-excited braking to function. Sensor noise and parameter drift are secondary concerns within realistic ranges.

Figure 13 provides a visual overview of all three sweeps.

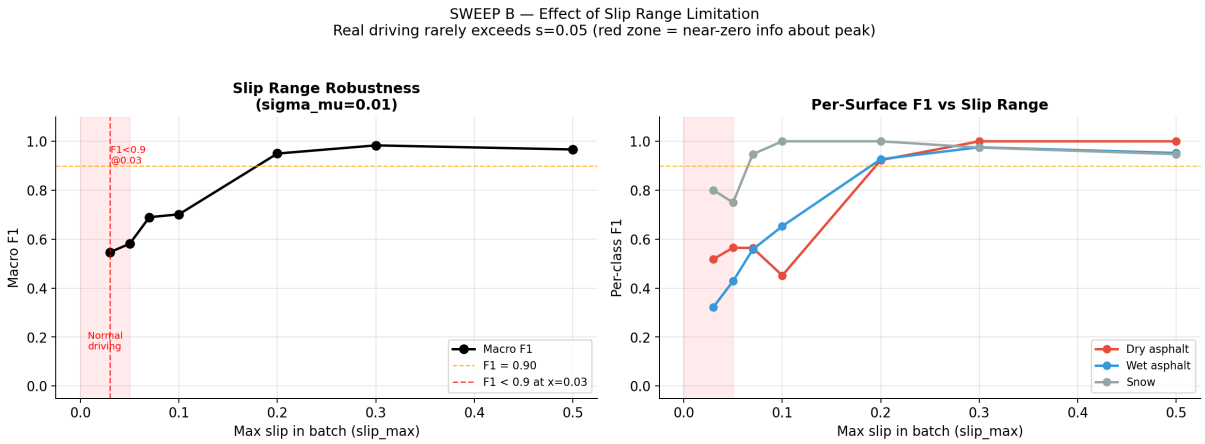

Figure 11: Sweep B: Macro-F_1 vs. maximum slip excitation in each batch ($\sigma_\mu = 0.01$, batch size 50). The red shaded region ($s < 0.05$) corresponds to normal driving without active ABS engagement. F_1 drops to 0.58 at $s_{\max} = 0.05$, confirming that the classifier requires ABS-excited slip to function reliably.

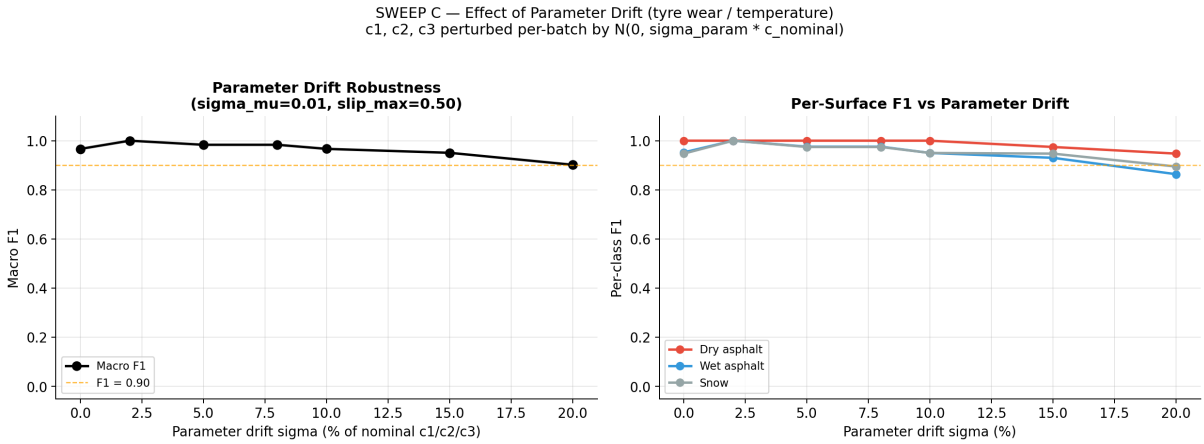

Figure 12: Sweep C: Macro-F_1 vs. relative parameter drift σ_p ($\sigma_\mu = 0.01$, $s_{\max} = 0.50$, batch size 50). The classifier remains above $F_1 = 0.90$ up to $\pm 20\%$ drift, confirming robustness to tyre wear and temperature variation.

Table 4: Robustness summary: failure mode, onset, and implication.

Failure mode	Parameter	$F_1 = 0.90$ threshold	F_1 at nominal	Implication
Sensor noise	σ_μ	~ 0.05	0.967 @ 0.01	Standard sensors sufficient
Slip limitation	$s_{\max}$	~ 0.20	0.967 @ 0.50	Requires ABS excitation
Parameter drift	σ_p	> 0.20	0.967 @ 0.0	Highly robust

† All F_1 values for Sweeps B and C evaluated at $\sigma_\mu = 0.01$ baseline noise; nominal slip range $s \in [0.002, 0.50]$.

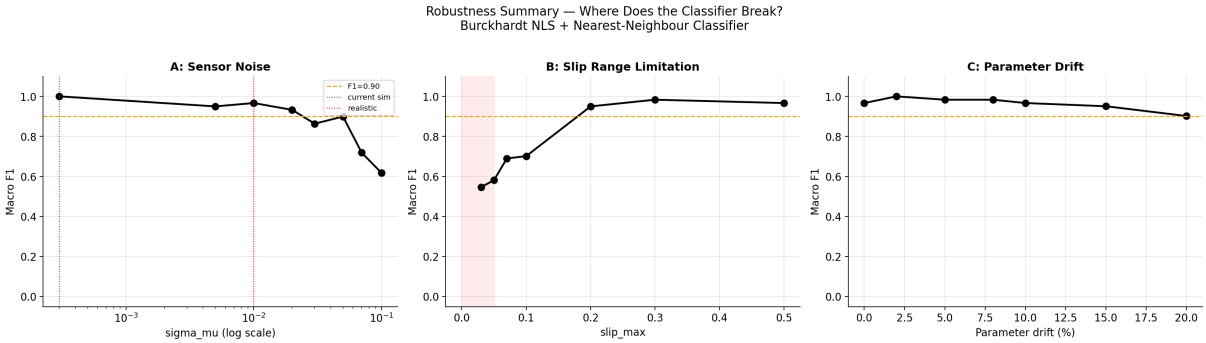

Figure 13: Robustness summary across all three sweeps. Left: sensor noise; centre: slip range; right: parameter drift. The orange dashed line marks $F_1 = 0.90$. Slip range limitation (centre) is the dominant failure mode.

7 Discussion

7.1 Operating Envelope

The results establish a clear operating envelope for the proposed classifier: it functions reliably (macro-$F_1 > 0.90$) when:

- Slip is excited beyond $s > 0.10$–0.20 (ABS or emergency braking),
- Friction sensor noise is below $\sigma_\mu \approx 0.05$, and
- Tyre parameter drift is within $\pm 20\%$ of nominal values.

These conditions are satisfied during ABS-controlled braking events. Outside this envelope—particularly during normal cruise driving ($s < 0.05$)—the classifier should be considered unreliable.

7.2 Limitations and Future Work

7.2.1 Real experimental data.

The most important next step is validation on real tire test-bench data (e.g., TU Delft flat-track rig [9], NHTSA braking datasets). The simulation evaluation, even with added noise, cannot substitute for real measurements due to model circularity.

7.2.2 Model mismatch.

All evaluations assumed the Burckhardt model is the correct generative model. Real tire friction deviates from Burckhardt due to hysteresis, load sensitivity, and rubber compound variation. Testing with a Pacejka Magic Formula generator and a Burckhardt fitter would break the circularity and provide a more honest assessment.

7.2.3 Surface transitions.

The current classifier assumes a single-surface batch. Transitions (e.g., wet patch on dry road mid-batch) are not handled. A sliding-window approach with change-point detection would be required.

7.2.4 Out-of-distribution surfaces.

The classifier has no rejection mechanism. Adding a confidence threshold—e.g., refusing to classify when $\min_k d_k > d_{\text{threshold}}$—would prevent false surface assignments on ice, gravel, or mixed surfaces.

7.3 Industrial Application

In the context of the NTT DATA BEST Hackathon, this work directly addresses **Use Case 1** (real-time friction identification, MAKE decision): the three-layer ESC identifier replaces a manual 240-hour annual calibration campaign with a continuously self-updating parameter estimation system. The offline NLS classifier serves as a validation and calibration-campaign analysis tool, not as the real-time surface detector (which is the ESC identifier itself).

8 Conclusion

We have presented the design and validation of a three-layer ESC-gradient Burckhardt identifier, implemented in Python from a MATLAB reference implementation. The identifier achieves rapid convergence on synthetic simulation data across dry asphalt, wet asphalt, and snow surfaces.

A rigorous evaluation framework exposes a fundamental limitation of closed-loop simulation evaluation: perfect classification metrics ($F_1 = 1.000$) are mathematically guaranteed when the test data is generated by the same model used for fitting. We explicitly characterise this as a software sanity check rather than a classifier accuracy result.

A three-source noise model introduced to simulate realistic automotive sensor conditions reveals:

1. The classifier achieves $F_1 = 0.967$ at realistic friction sensor noise ($\sigma_\mu = 0.01$), remaining above 0.90 up to $\sigma_\mu = 0.05$.
2. Slip range limitation is the dominant failure mode: F_1 drops to 0.58 for $s < 0.05$, confirming that ABS excitation is a prerequisite.
3. The classifier is highly robust to parameter drift ($F_1 > 0.90$ for up to $\pm 20\%$ variation), because the three standard surfaces are well-separated in Burckhardt parameter space.

This work provides a fully reproducible baseline for future validation on real experimental data, with all code and the MATLAB noise injection script publicly available in the companion repository.

Code and Data Availability

All Python source code, the MATLAB export scripts, the preprocessing pipeline, the three-source noise model (`eval_robustness.py`), and the classification evaluation scripts are available at:

https://github.com/Sajjad-Shahali/NTT_DATA_hackathon

The Simulink simulation data (`friction_data_full.csv`) can be regenerated by running `run_validation_updated.m` followed by `export_for_python.m` in MATLAB with the ESC framework installed. The noisy variant is produced by `export_for_python_noisy.m`, with noise parameters configurable at the top of the script.

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Author Contributions

All team members contributed equally to the design, implementation, and validation of the system. Algorithm analysis and Python port: all authors. Simulation data generation: MATLAB team members. Machine-learning evaluation and robustness analysis: all authors. Report writing: all authors.

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